

6.4 Technical Processes

6.4.1 Stakeholder Requirements Definition Process

This International Standard specializes the Stakeholder Requirements Definition Process of ISO/IEC 15288. Users may claim conformance to the 15288 process where appropriate. The process can be tailored according to organizational policies, application domains, and project contexts.

6.4.1.1 Purpose

The purpose of this process is to define the requirements for a system that provides services to users and other stakeholders in a defined environment. It identifies stakeholders, their needs, and stakeholder classes throughout the system life cycle. It analyzes and transforms these needs into a common set of stakeholder requirements that:

- Express the system's intended interaction with its environment;
 - Serve as a reference for validating each resulting operational service;
- and
- Provide a basis for defining system requirements in 6.4.2.
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6.4.1.2 Outcomes

As a result of successful implementation of the Stakeholder Requirements Definition Process:

- a) The required characteristics and context of use of services are specified.
 - b) Constraints on a system solution are defined.
 - c) Traceability of stakeholder requirements to stakeholders and their needs is achieved.
 - d) The basis for defining system requirements is described.
 - e) The basis for validating the conformance of services is defined.
 - f) A basis for negotiating and agreeing to supply a service or product is provided.
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6.4.1.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.1.3.1 Stakeholder Identification

This activity identifies all relevant stakeholders and stakeholder classes.

- 6.4.1.3.1.1 Identify individual stakeholders or stakeholder classes with legitimate interests in the system throughout its life cycle.
 - 6.4.1.3.1.2 Include users, operators, supporters, developers, producers, trainers, maintainers, disposers, acquirers and suppliers, parties responsible for external interfacing entities, regulatory bodies, and members of society.
 - 6.4.1.3.1.3 Where direct communication is not practicable, select representatives or designated proxy stakeholders.
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6.4.1.3.2 Requirements Identification

This activity elicits, analyzes, and documents stakeholder requirements and constraints.

- 6.4.1.3.2.1 Elicit stakeholder requirements using appropriate methods (e.g., interviews, workshops, surveys, models).
- 6.4.1.3.2.2 Use product quality models and quality requirements (e.g., ISO/IEC 9126-1, ISO/IEC 25030) to guide identification of quality characteristics.
- 6.4.1.3.2.3 Include societal needs, acquiring organization constraints, and user/operator capabilities and operational characteristics.
- 6.4.1.3.2.4 Cite requirement sources, including stakeholder assumptions and the importance placed on satisfying each requirement.
- 6.4.1.3.2.5 Define measures of effectiveness for key stakeholder needs to assess operational performance.
- 6.4.1.3.2.6 Identify significant human-system interaction risks across the system life cycle. Refer to ISO PAS 18152 for guidance on identifying and treating human factors issues.
- 6.4.1.3.2.7 Define unavoidable constraints on the system solution due to existing agreements, management decisions, and technical decisions.
- 6.4.1.3.2.8 Define representative sequences of activities to identify all required services for anticipated operational and support scenarios and environments, including unstated legal, regulatory, and social requirements.
- 6.4.1.3.2.9 Identify user-system interactions, considering human capabilities and limitations (physical, mental, learned skills), workplace conditions, environments, equipment, and scenarios (normal, unusual, emergency).

- 6.4.1.3.2.10 Specify health, safety, security, environmental, and other stakeholder requirements.
 - If usability is important, plan, specify, and implement usability requirements throughout the life cycle, referencing ISO 9241-11, ISO 13407, and ISO/TR 18529.
 - Address potential adverse effects on human health and safety for critical qualities.
 - Identify safety risks and specify safety requirements and functions, referencing applicable standards (e.g., IEC 61508) and accepted professional practices.
 - Identify security risks and specify security functions across physical, procedural, communications, computing, data, and emissions domains. Include access control, information protection, and denial-of-service considerations.
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6.4.1.3.3 Requirements Evaluation

This activity analyzes the set of elicited requirements to ensure they are suitable for further system definition.

- 6.4.1.3.3.1 Analyze the complete set of stakeholder requirements to detect conflicts, omissions, incompleteness, ambiguity, inconsistency, incongruity, or unverifiability.
 - 6.4.1.3.3.2 Use appropriate evaluation techniques, such as inspections, reviews, model checks, or requirements quality checklists.
 - 6.4.1.3.3.3 Prioritize and resolve issues to produce a consistent, complete, and feasible set of stakeholder requirements.
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6.4.1.3.4 Requirements Agreement

This activity ensures stakeholder requirements are agreed upon and accepted.

- 6.4.1.3.4.1 Resolve problems with requirements that are conflicting, impractical, or unrealizable.
 - 6.4.1.3.4.2 Feed back analyzed requirements to stakeholders to verify that needs and expectations are adequately captured and expressed.
 - 6.4.1.3.4.3 Present and obtain agreement on proposals to resolve conflicting or impractical requirements.
 - 6.4.1.3.4.4 Confirm with stakeholders that requirements are expressed correctly and fully represent their needs.
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6.4.1.3.5 Requirements Recording

This activity records and maintains stakeholder requirements and traceability.

- 6.4.1.3.5.1 Record stakeholder requirements in an appropriate format to establish baselines, retain changes, and provide traceability to system requirements.
- 6.4.1.3.5.2 Maintain traceability between stakeholder requirements and their sources, ensuring changes in needs are identified and reflected at key decision points.
- 6.4.1.3.5.3 Use these records as a knowledge base for subsequent system development and for communicating requirement status to stakeholders.

Cross-References

- **System Requirements Analysis Process (6.4.2)** — for transforming stakeholder requirements into system requirements.
- **Validation Process (7.2.5)** — for validating that the delivered system satisfies stakeholder requirements.
- **Decision Management (6.3.3)** and **Configuration Management (6.3.4)** — for managing requirement conflicts and baselines.